Journal Club

Grokking as the Transition from Lazy to Rich Training Dynamics

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LYH @ PMI group meeting

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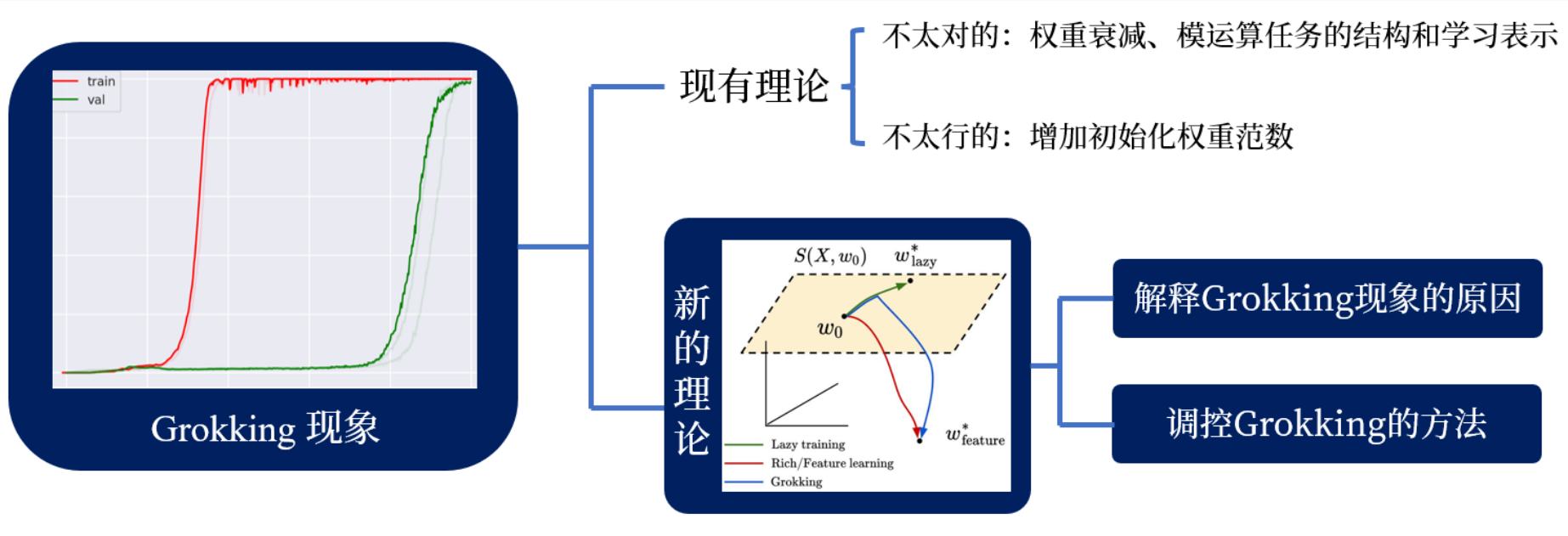
Generalization



Q&I: “顿悟”现象，大模型的涌现

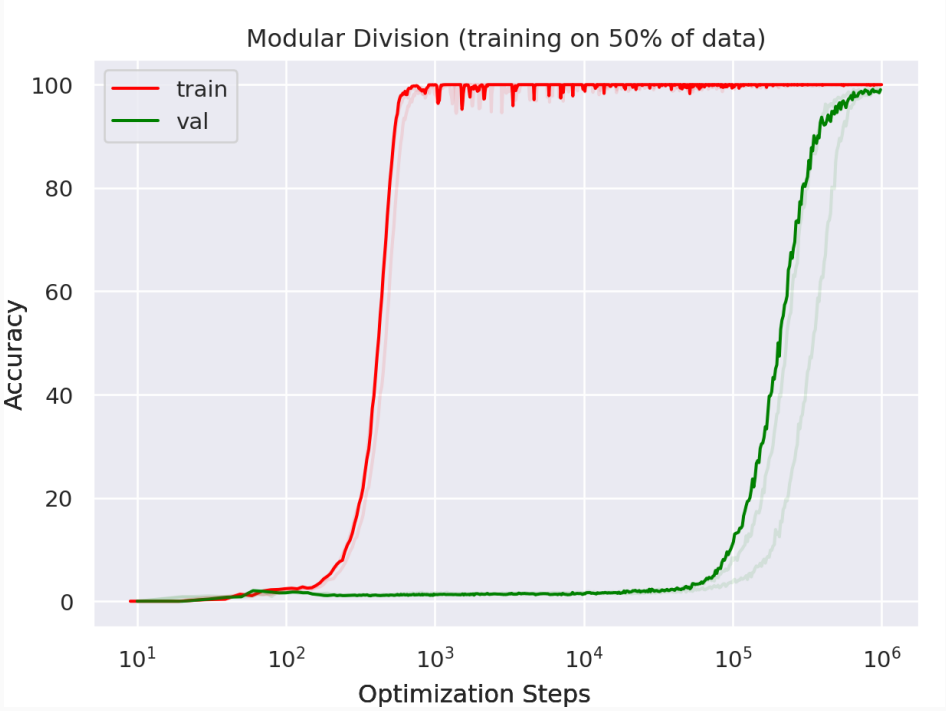
Why: 作者觉得以前的解释不太行

How: 基于“高维多项式回归任务”做的一系列实验验证

什么是 Grokking 现象?

Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv: 2201.02177*, 2022.

模运算任务 $a + b = c \bmod p$



现有的研究:

- 学习过程中权重衰减
- 初始化时改变权重的范数
- 模运算任务的结构和学习表示

...

批判大会：二层感知机中的模运算任务

模型设定：

老师学生网络

激活函数为 tanh

$p = 23$

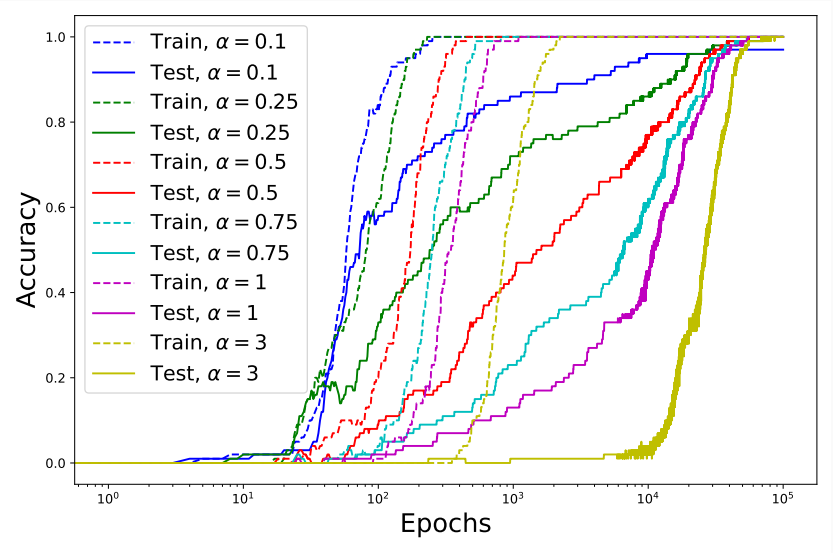
输入层 $D = 2p$

隐藏层 $N = 100$

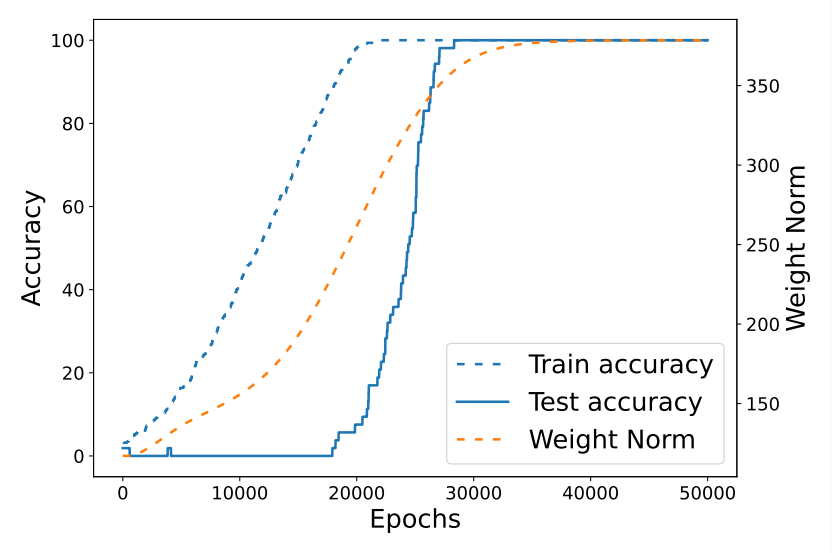
学习率 $\eta = 100$

训练集 90%，测试集 10%

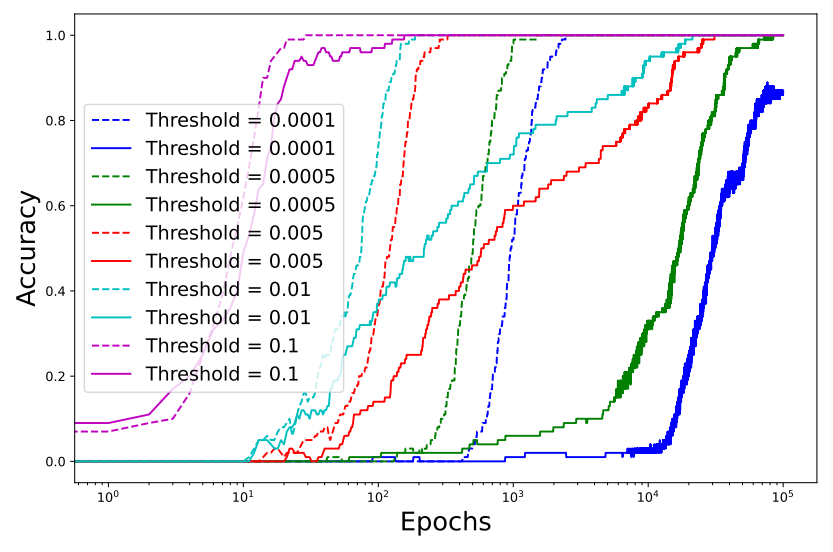
2. 初始化时加倍权重 vs 缩放参数 α



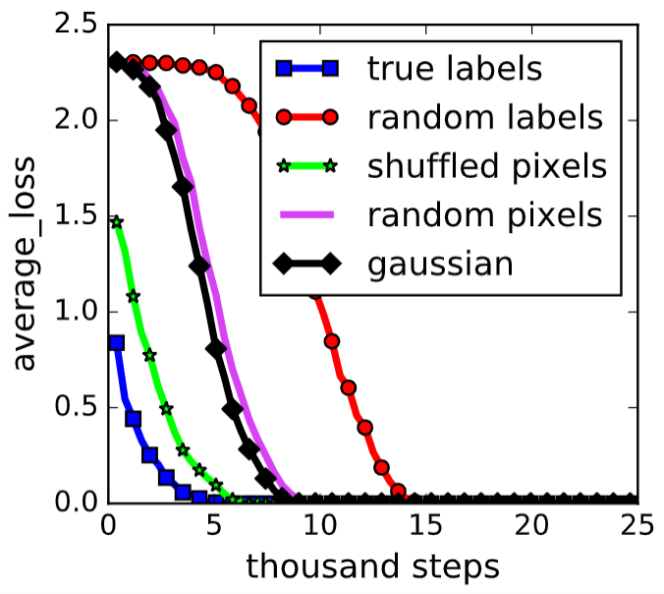
1. 学习过程中的权重衰减



3. 准确率曲线不靠谱 threshold



Lazy Learning



网络学习随机标签

$\Downarrow$

网络过度参数化

$\Downarrow$

缩放因子

$\Downarrow$

Lazy Learning

- Lazy Learning 的本质：

网络在初始化时可以近似为线性模型

$$f(\boldsymbol{w}, \boldsymbol{x}) \approx f(\boldsymbol{w}_0, \boldsymbol{x}) + \nabla_{\boldsymbol{w}} f(\boldsymbol{w}, \boldsymbol{x})|_{\boldsymbol{w}_0} \cdot (\boldsymbol{w} - \boldsymbol{w}_0)$$

- 逼近线性化极限的方式：

1. Large width
2. Large initial weights
3. Label rescaling α^{-1}
4. Output rescaling α

- 核方法：$K(\boldsymbol{x}_i, \boldsymbol{x}_j) = \langle \phi(\boldsymbol{x}_i), \phi(\boldsymbol{x}_j) \rangle$

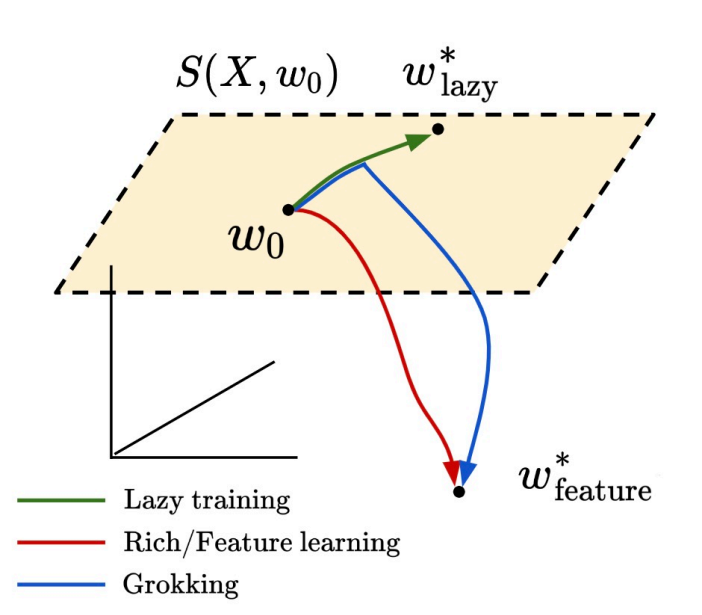
无限宽神经网络 $\Leftrightarrow$ 神经正切核

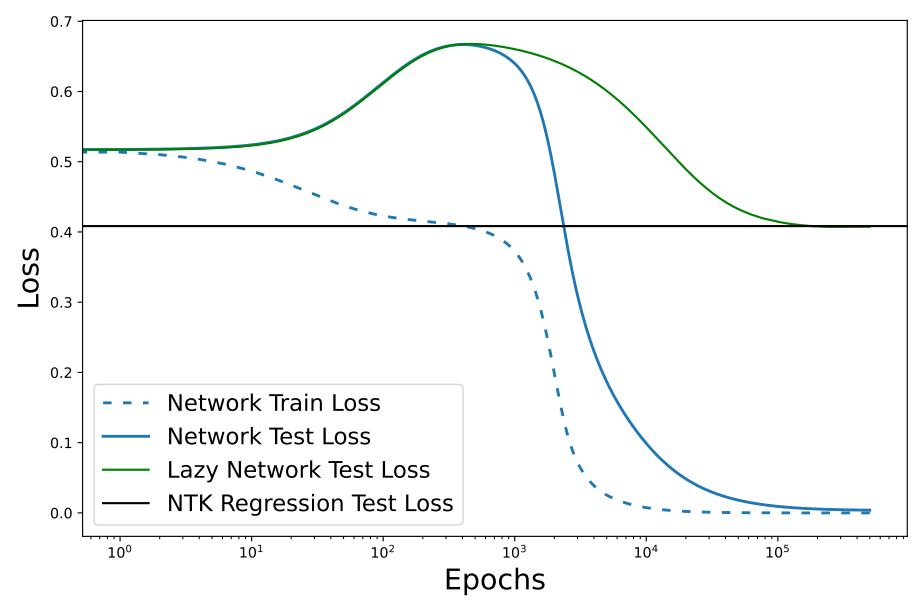
$$NTK(\boldsymbol{x}_i, \boldsymbol{x}_j) = \boldsymbol{x}_i^\top \boldsymbol{x}_j \langle \sigma'(\boldsymbol{x}_i), \sigma'(\boldsymbol{x}_j) \rangle$$

核函数与目标函数的对齐

作者的观点：*Grokking as Transition from Lazy to Rich Learning*

$S(X, w_0)$： $\boldsymbol{w}_0 + \mathrm{span}\{\nabla_{\boldsymbol{w}} f(\boldsymbol{w}, \boldsymbol{x}_\mu)|_{w_0}\}_{\mu=1}^{P}$ 构成的仿射子空间





影响动力学转变的关键参数：

- **网络输出的缩放因子 α**：网络偏离线性化的程度，即特征学习的速度
- **初始化的核函数与目标函数是否对齐**：泛化能力和特征学习的多少

多项式回归任务

输入 $\boldsymbol{x} \in \mathbb{R}^D \sim \mathcal{N}(0, \frac{1}{D}\boldsymbol{I})$ **标签** $y(\boldsymbol{x}) = \frac{1}{2}(\boldsymbol{\beta}_\star \cdot \boldsymbol{x})^2$ **训练集** $\{(\boldsymbol{x}_\mu, y_\mu)\}_{\mu=1}^P$

二层感知机模型 $f(\boldsymbol{w}, \boldsymbol{x}) = \frac{\alpha}{N}\sum_{i=1}^N \phi(\boldsymbol{w}_i \cdot \boldsymbol{x})$ 其中 $\phi(h) = h + \frac{\epsilon}{2}h^2$

$$f(\boldsymbol{w}, \boldsymbol{x}) = \frac{\alpha}{N}\sum_{i=1}^N \boldsymbol{w}_i\boldsymbol{x} + \frac{\alpha}{N}\sum_{i=1}^N \frac{\epsilon}{2}(\boldsymbol{w}_i^\top \boldsymbol{x})^\top(\boldsymbol{w}_i^\top \boldsymbol{x}) = \frac{\alpha}{N}\sum_{i=1}^N \boldsymbol{w}_i\boldsymbol{x} + \frac{\alpha}{N}\sum_{i=1}^N \frac{\epsilon}{2}\boldsymbol{x}^\top \boldsymbol{w}_i\boldsymbol{w}_i^\top \boldsymbol{x}$$

定义 $\bar{\boldsymbol{w}} = \dfrac{1}{N}\sum_{i=1}^N \boldsymbol{w}_i \quad \boldsymbol{M} = \dfrac{1}{N}\sum_{i=1}^N \boldsymbol{w}_i\boldsymbol{w}_i^\top$，重写为 $f(\boldsymbol{x}) = \alpha\bar{\boldsymbol{w}}\boldsymbol{x} + \dfrac{\alpha\epsilon}{2}\boldsymbol{x}^\top \boldsymbol{M}\boldsymbol{x}$

NTK 模型

$$\begin{aligned}
K(\boldsymbol{x}, \boldsymbol{x}') &= \boldsymbol{x} \cdot \boldsymbol{x}' \left\langle \phi'\left(\boldsymbol{w}_i \cdot \boldsymbol{x}\right), \phi'\left(\boldsymbol{w}_i \cdot \boldsymbol{x}'\right)\right\rangle \\
&= \boldsymbol{x} \cdot \boldsymbol{x}' \left\langle \left(1 + \epsilon\boldsymbol{w}_i \cdot \boldsymbol{x}\right), \left(1 + \epsilon\boldsymbol{w}_i \cdot \boldsymbol{x}'\right)\right\rangle \\
&= \boldsymbol{x} \cdot \boldsymbol{x}' + \epsilon(\boldsymbol{x} \cdot \boldsymbol{x}')\ \bar{\boldsymbol{w}} \cdot (\boldsymbol{x} + \boldsymbol{x}') + \epsilon^2(\boldsymbol{x} \cdot \boldsymbol{x}')\ \boldsymbol{x}^\top \boldsymbol{M}\boldsymbol{x}'
\end{aligned}$$

多项式回归任务

为什么选这个任务？

1. 对于 D 维 k 次多项式回归，二层感知机学习需要的数据量 $P \sim D$

2. 核回归任务中，达到良好的泛化能力需要的数据量 $P \sim \lambda^{-1}$

其中 λ 是本征值 $\int \mathrm{d}\boldsymbol{x}\ p(\boldsymbol{x})\ K(\boldsymbol{x}, \boldsymbol{x}')\phi(\boldsymbol{x}) = \lambda\phi(\boldsymbol{x}')$

- 线性目标函数 $\lambda = D^{-1}$，$P \sim D$
- 本任务中，标签可以改写为

$$y(\boldsymbol{x}) = \frac{1}{2}\left[(\boldsymbol{x}\cdot\boldsymbol{\beta}_\star)^2 - \frac{1}{D}|\boldsymbol{x}|^2\right] + \frac{1}{2D}|\boldsymbol{x}|^2$$

第一项 $\lambda = 2\epsilon^2 D^{-2}$，$P \approx D^2$，第二项 $\lambda = \frac{1}{D}\left(1 + \frac{2}{D}\right)$，$P \approx D$

This separation of timescales motivated the original investigation of this task in which to find grokking.

多项式回归任务

ϵ 到底控制了啥？ ϕ 与 y 的对齐程度

$$y(\boldsymbol{x}) = \frac{1}{2}(\boldsymbol{\beta}_\star \cdot \boldsymbol{x})^2 \qquad \phi(\boldsymbol{w}_i\boldsymbol{x}) = \boldsymbol{w}_i\boldsymbol{x} + \frac{\epsilon}{2}(\boldsymbol{w}_i\boldsymbol{x})^2$$

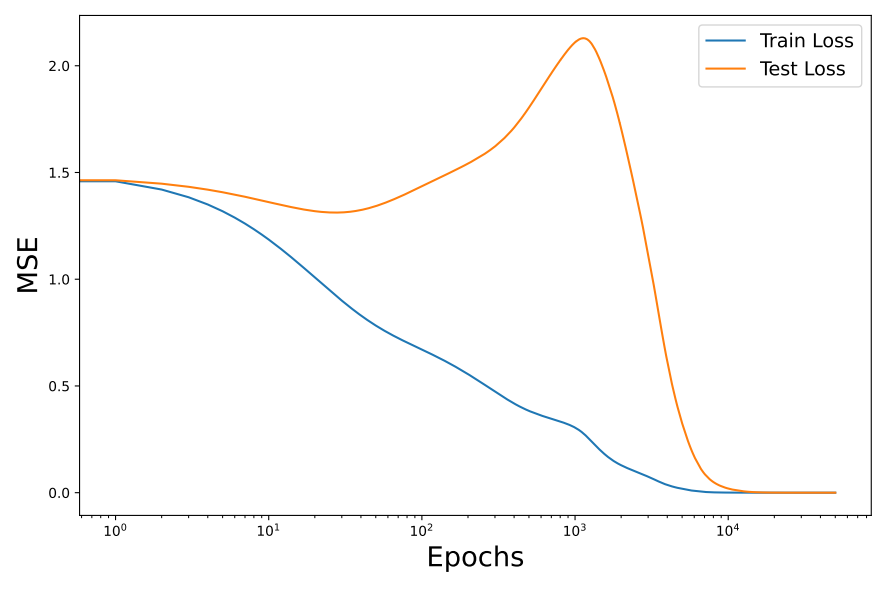
Loss 分解

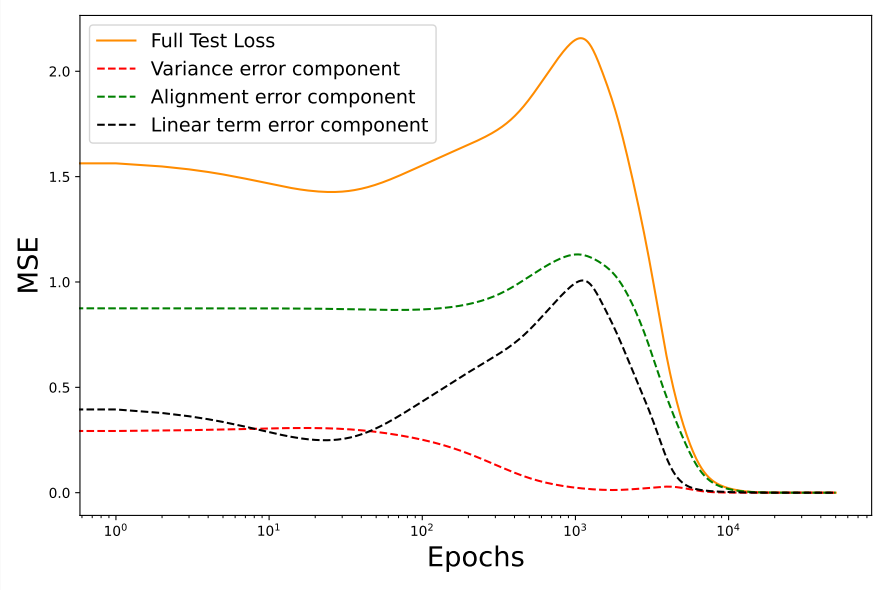
$$\frac{1}{DN}\sum_i |\boldsymbol{w}_i|^2 = \frac{1}{D}\sum_{j=1}^{D} M_{jj}$$

$$\mathcal{L} = \langle (y-f)^2 \rangle$$

$$= \underbrace{\frac{1}{4}\left(\frac{1}{D}|\boldsymbol{\beta}_\star|^2 - \frac{\alpha\epsilon}{D}\mathrm{Tr}\boldsymbol{M}\right)^2}_{\text{variance error}} + \underbrace{\frac{1}{2D^2}\left|\alpha\epsilon\boldsymbol{M} - \boldsymbol{\beta}_\star\boldsymbol{\beta}_\star^\top\right|_F^2}_{\text{misalignment error}} + \underbrace{\frac{\alpha^2}{D}|\bar{\boldsymbol{w}}|^2}_{\text{power in linear modes}}$$

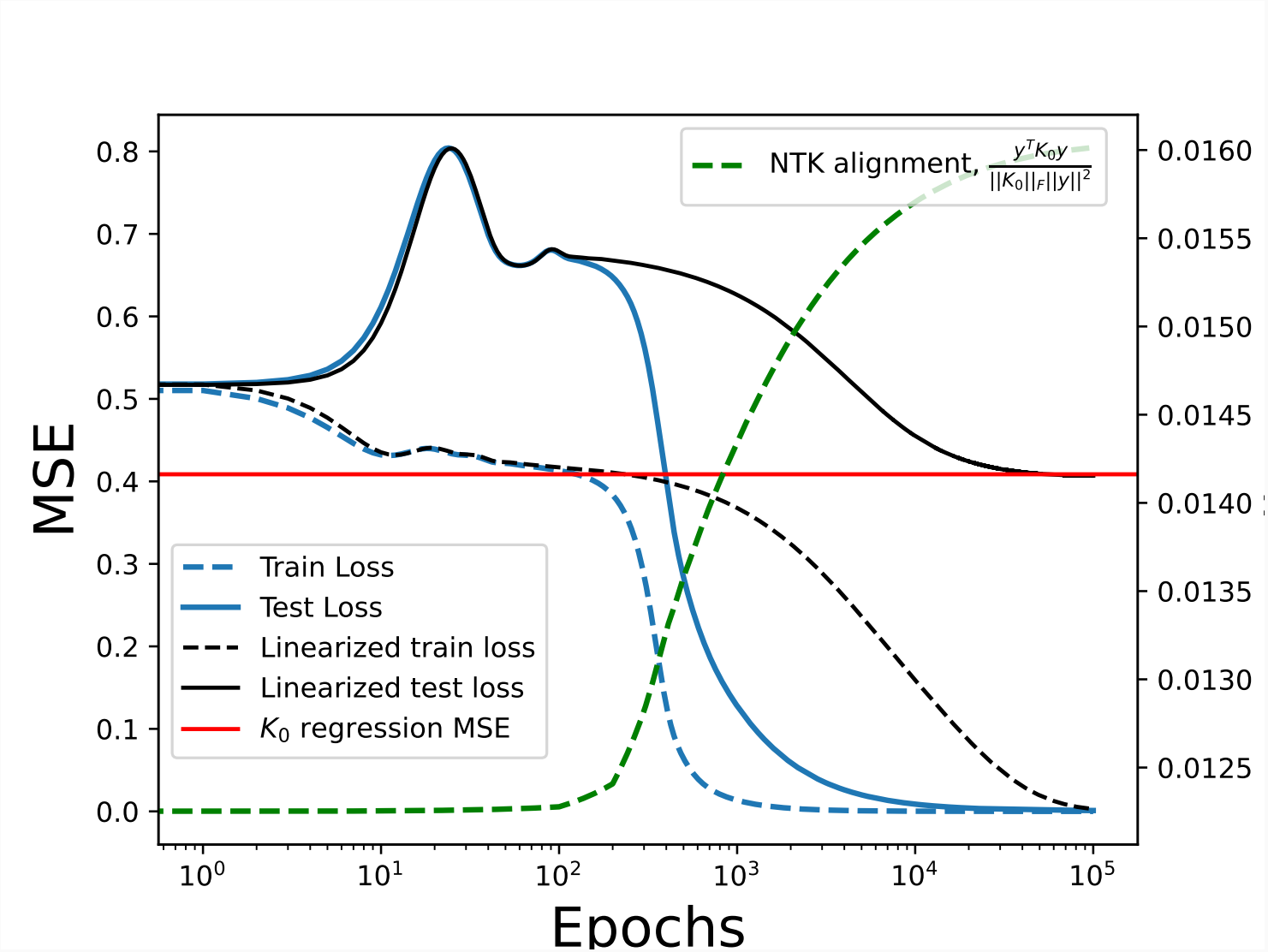
没对齐的时候线性项会导致 Grokking 的出现



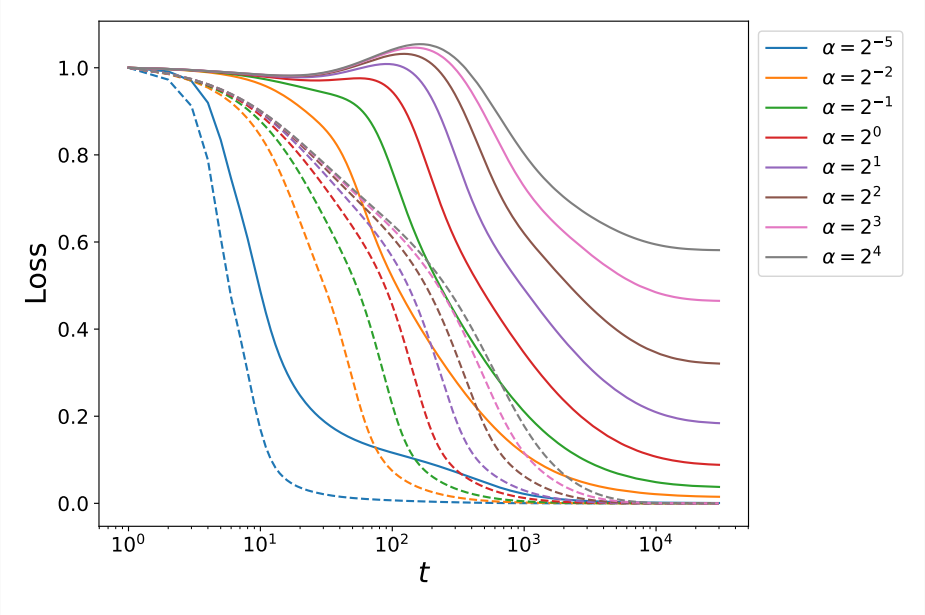


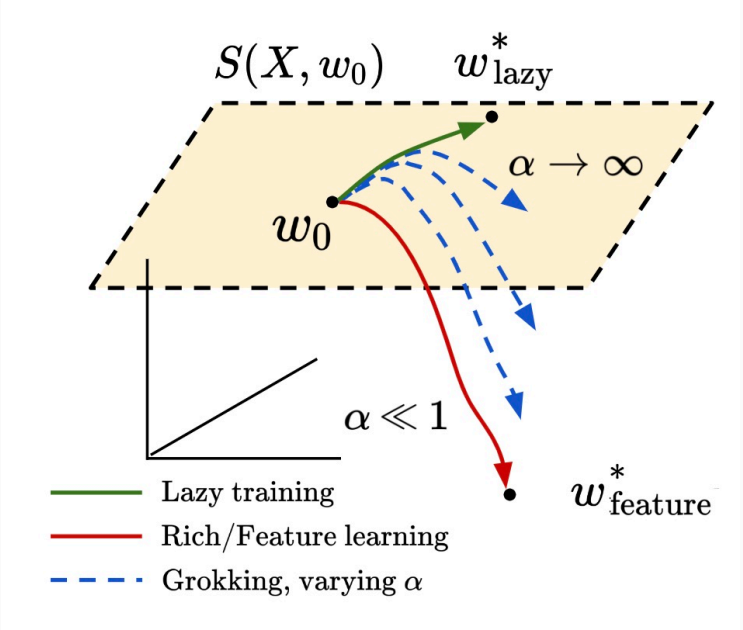
先从后面借个图

centered-kernel alignment (CKA) $\frac{y^T K_0 y}{||K||_F ||y||^2}$ 越大表明 ϕ 与 y 对齐的越好



实验结果　改变 α 控制 *Feature Learning* 的速度

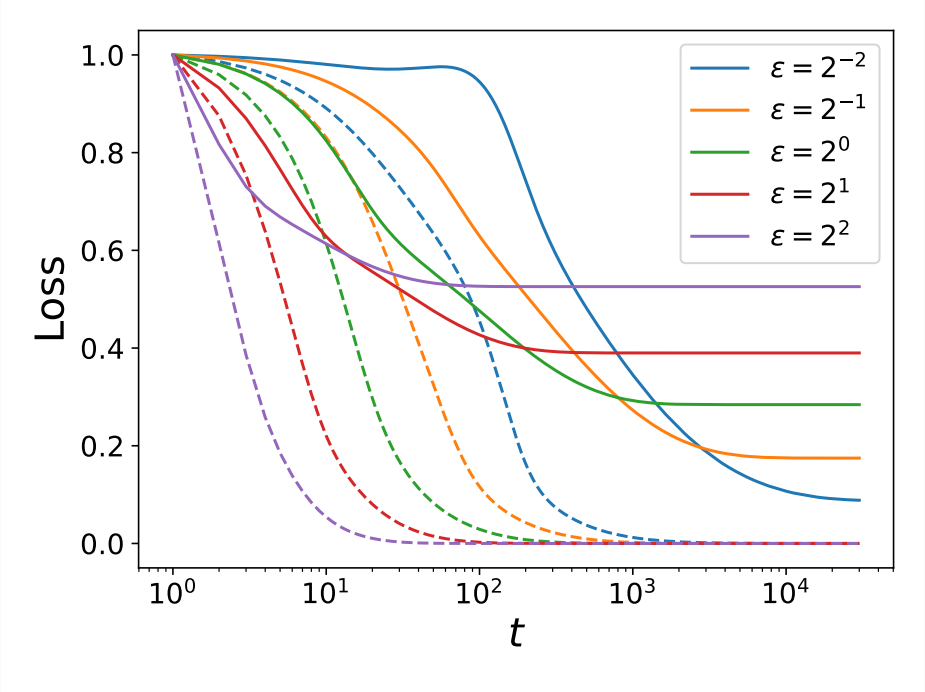


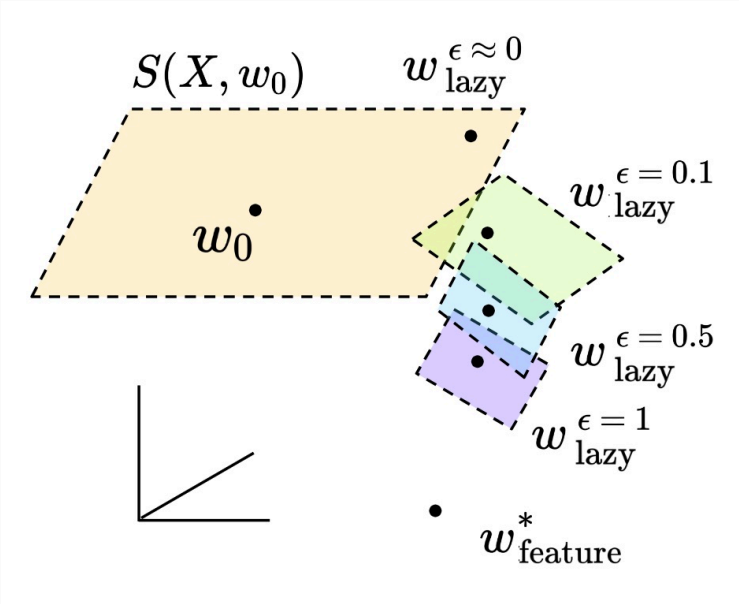


实验结果　改变 ϵ 控制需要学习多少特征

ϵ 越小，对齐的越差，NTK 的泛化越差，特征学习的更多

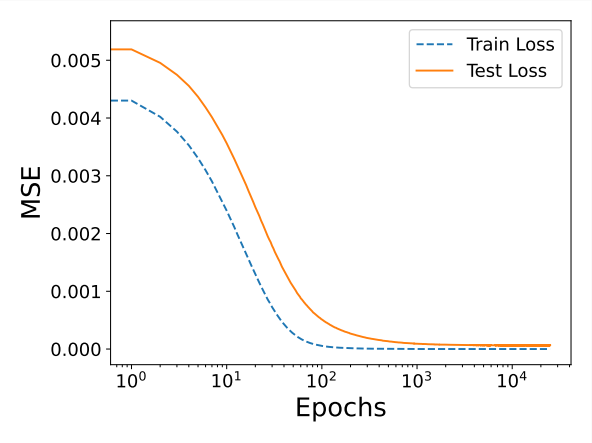
ϵ 越大，对齐的越好，NTK 的泛化越好，特征学习的越少

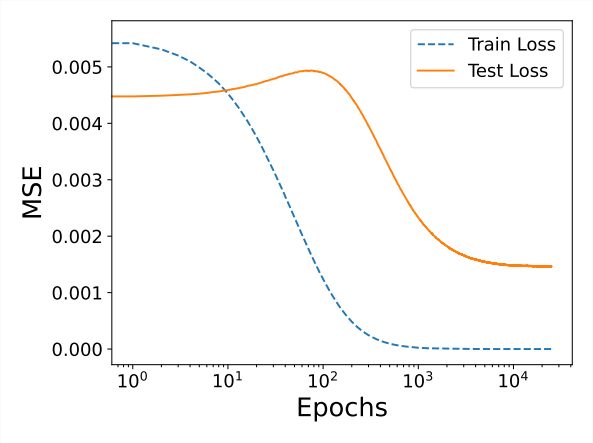


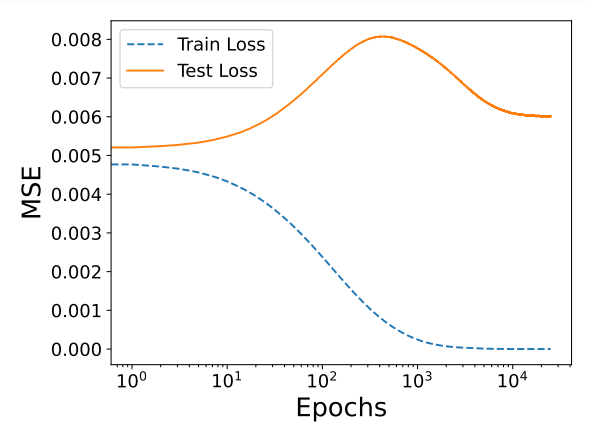


实验结果　其它调整对齐的方式

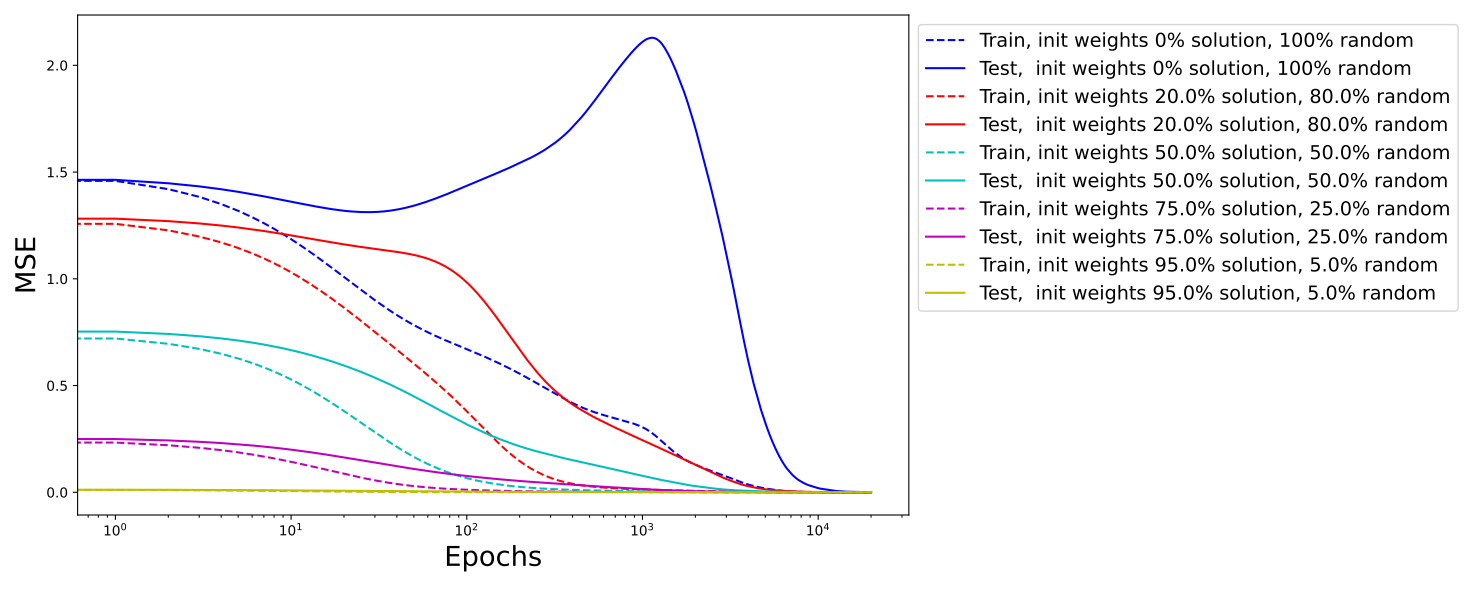
将任务标签替换为初始化 NTK 核的特征向量，特征值从大到小，对齐程度从好到差



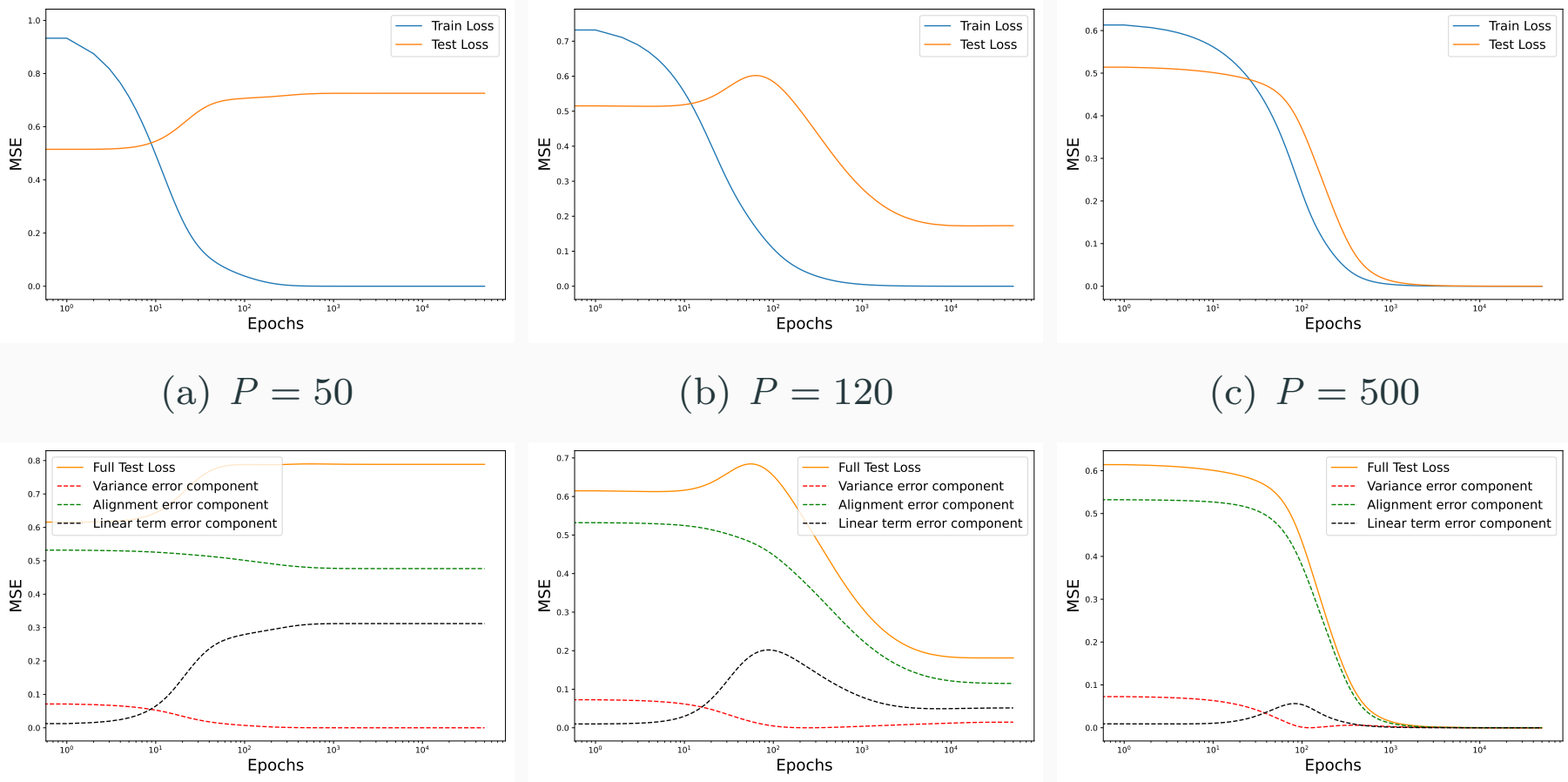




从随机权重到对齐解的插值



实验结果　改变数据集大小



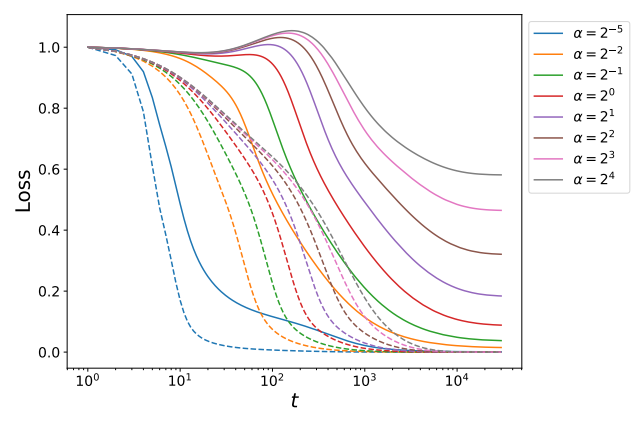
(a) $P = 50$　(b) $P = 120$　(c) $P = 500$

能否成功泛化的关键在于能否把初始化时的线性项学习掉

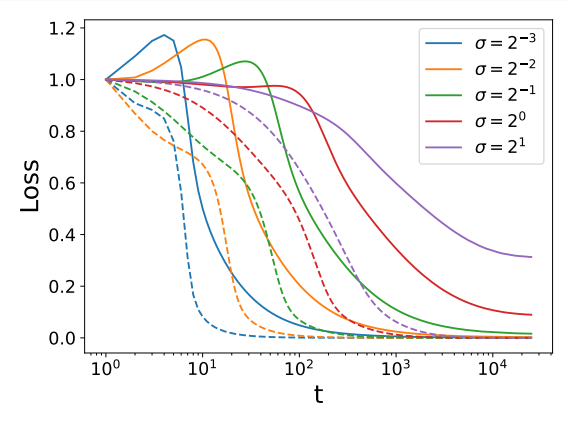
怎么搞定线性项？增大数据量！

再来解释以前的工作

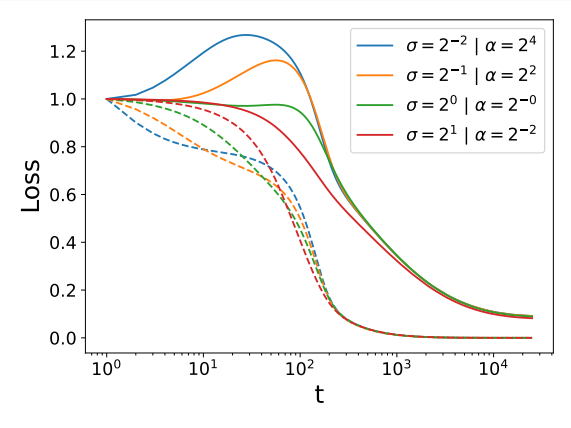
初始化权重范数



(a) 改变 α

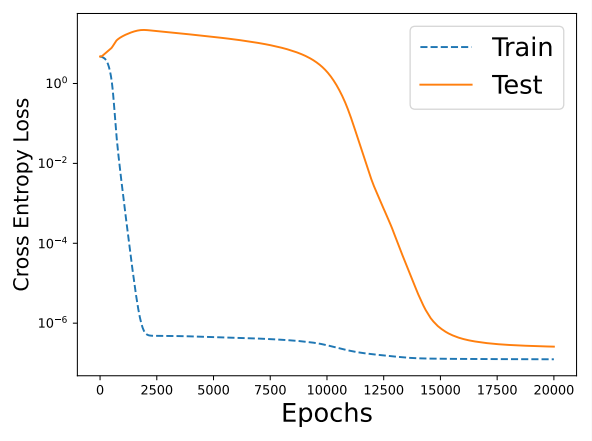


(b) 改变权重初始化尺度 σ

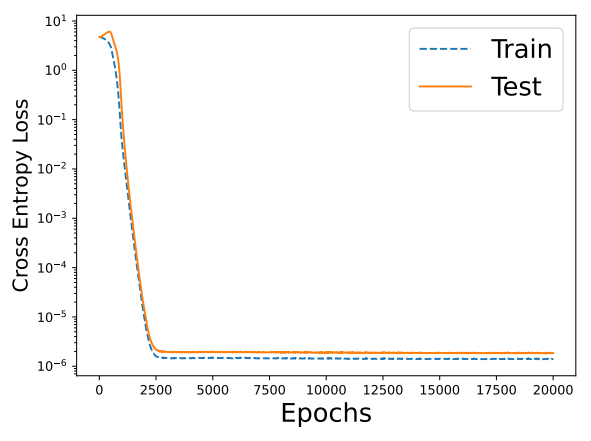


(c) 固定 $\sigma^2\alpha$ 保持特征学习

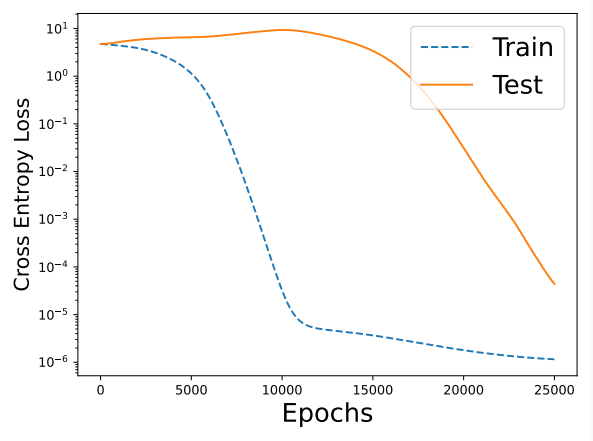
权重衰减（迫使网络离开非线性状态）



(d) $\alpha = 0.5$, wd=1



(e) $\alpha = 0.5$, wd=5



(f) $\alpha = 1.4$, wd=5

Summary

结论：

1. Grokking 是因为网络先 Lazy Learning 然后转变到 Feature Learning

- 为什么一开始是 Lazy Learning? 网络初始化时近似线性
- 为什么 Lazy Learning 时没有泛化? 初始化的时候 NTK 和目标没对齐
- 什么时候能够泛化? 没对齐导致的 Loss 减小

2. 怎么调控 Grokking 现象？

- 改 α，让网络尽快脱离线性化
- 改 ϵ，把初始化和任务对齐，立即泛化

局限性：

- 没有搞定从记忆转变到泛化需要的样本大小和训练时间的理论
- AdamW 优化器在产生 Grokking 中有什么作用?

吐槽： 实验做的很丰富，理论推导不多而且写的一般，文章组织的很烂